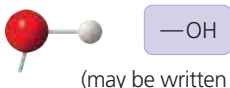
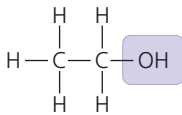
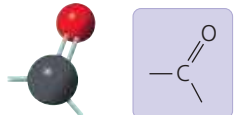
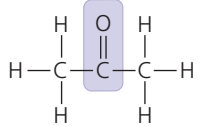
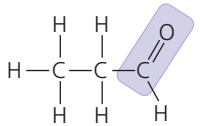
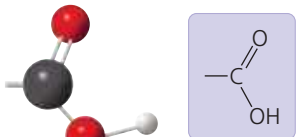
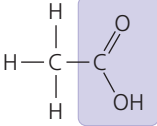
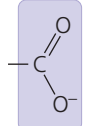
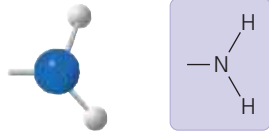
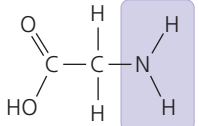
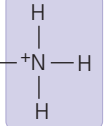
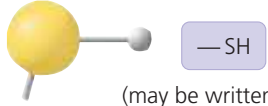
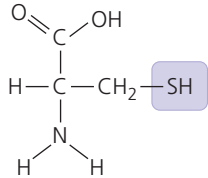
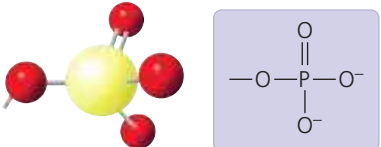
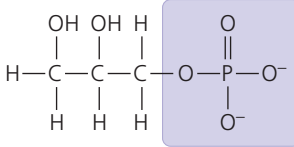
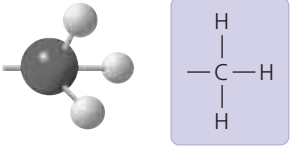
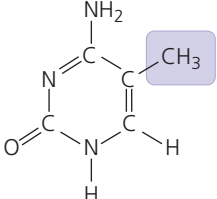


▼ **Figure 4.9** Some biologically important chemical groups.

| Chemical Group                                                                                                                                                    | Group Properties and Compound Name                                                                                                                                                                                                                                                                                                | Examples                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hydroxyl group</b> ( $\text{—OH}$ )<br><br>(may be written $\text{HO—}$ )     | Is polar due to electronegative oxygen. Forms hydrogen bonds with water, helping dissolve compounds such as sugars.<br><br>Compound name: <b>Alcohol</b> (specific name usually ends in <i>-ol</i> )                                                                                                                              |  <b>Ethanol</b> , the alcohol present in alcoholic beverages                                                                                                                                                                                        |
| <b>Carbonyl group</b> ( $\text{>C=O}$ )<br>                                      | Sugars with ketone groups are called ketoses; those with aldehydes are called aldoses.<br><br>Compound name: <b>Ketone</b> (carbonyl group is within a carbon skeleton) or <b>aldehyde</b> (carbonyl group is at the end of a carbon skeleton)                                                                                    |  <b>Acetone</b> , the simplest ketone  <b>Propanal</b> , an aldehyde                                                                                             |
| <b>Carboxyl group</b> ( $\text{—COOH}$ )<br>                                     | Acts as an acid (can donate $\text{H}^+$ ) because the covalent bond between oxygen and hydrogen is so polar.<br><br>Compound name: <b>Carboxylic acid</b> , or <b>organic acid</b>                                                                                                                                               |  <b>Acetic acid</b> , which gives vinegar its sour taste $\rightleftharpoons$  $+\text{H}^+$<br>Ionized form of $\text{—COOH}$ (carboxylate ion), found in cells |
| <b>Amino group</b> ( $\text{—NH}_2$ )<br>                                       | Acts as a base; can pick up an $\text{H}^+$ from the surrounding solution (water, in living organisms).<br><br>Compound name: <b>Amine</b>                                                                                                                                                                                        |  $+\text{H}^+$ $\rightleftharpoons$  <b>Glycine</b> , an amino acid (note its carboxyl group)<br>Ionized form of $\text{—NH}_2$ , found in cells               |
| <b>Sulfhydryl group</b> ( $\text{—SH}$ )<br><br>(may be written $\text{HS—}$ ) | Two $\text{—SH}$ groups can react, forming a “cross-link” that helps stabilize protein structure. Hair protein cross-links maintain the straightness or curliness of hair; in hair salons, “permanent” treatments break cross-links, then re-form them while the hair is in the desired shape.<br><br>Compound name: <b>Thiol</b> |  <b>Cysteine</b> , a sulfur-containing amino acid                                                                                                                                                                                                 |
| <b>Phosphate group</b> ( $\text{—OPO}_3^{2-}$ )<br>                            | Contributes negative charge (1– when positioned inside a chain of phosphates; 2– when at the end). When attached, confers on a molecule the ability to react with water, releasing energy.<br><br>Compound name: <b>Organic phosphate</b>                                                                                         |  <b>Glycerol phosphate</b> , which takes part in many important chemical reactions in cells                                                                                                                                                       |
| <b>Methyl group</b> ( $\text{—CH}_3$ )<br>                                     | Affects the expression of genes when bonded to DNA or to proteins that bind to DNA. Affects the shape and function of male and female sex hormones.<br><br>Compound name: <b>Methylated compound</b>                                                                                                                              |  <b>5-Methylcytosine</b> : Cytosine, a component of DNA, has been modified by addition of a methyl group.                                                                                                                                         |